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RESEARCH ARTICLE

Exploring Road Traffic Accidents Hotspots Using Clustering Algorithms and GIS-Based Spatial Analysis

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ABSTRACT This study conducts a comprehensive spatial analysis of road traffic accidents (RTAs) in Najran, a city emblematic of rapid urbanization in Saudi Arabia, which is facing significant public safety challenges due to an increase in vehicular traffic. By means of a dataset from 2022, we explore the spatial distribution of RTAs across the city's districts by employing advanced clustering algorithms, including Density-based Spatial Clustering of Applications with Noise (DBSCAN) and Hierarchical Agglomerative Clustering (HAC), as well as GIS-based density analysis, proximity analysis, and spatial interpolation, to unveil accident hotspots and disparities in emergency service coverage. Our findings reveal that (1) the HAC model, based on the Silhouette and Calinski-Harabasz Scores, performs better in identifying accident hotspots; (2) significant concentrations of accidents are observed along major highways and arterial roads, pinpointing critical hotspots within the city's fabric; (3) proximity analysis indicates gaps in the coverage of ambulance services and public hospitals relative to high-incident areas; (4) through spatial interpolation, detailed visualizations of RTA distributions are provided, revealing diverse accident patterns across Najran. The study highlights the critical role of spatial analysis in identifying high-risk areas and provides valuable insights for transport planners and public safety officials, supporting the development of targeted strategies to improve road safety and enhance emergency service responses.

INDEX TERMS Road traffic accidents, spatial analysis, Najran, Saudi Arabia.

I. INTRODUCTION

In an era marked by the critical role of vehicles in modern civilization, providing unmatched mobility and connectivity, these advancements also present significant challenges to public safety. The World Health Organization (WHO) highlights the serious issue of Road Traffic Accidents (RTAs) [1], particularly noting that the Eastern Mediterranean Region (EMR) has the second highest traffic casualty rate globally, surpassed only by the African Region [2]. This alarming statistic points to a widespread problem: RTAs emerge as a major public health concern, causing substantial economic, environmental, and social impacts worldwide [3].

In this setting, Najran, a swiftly growing city in Saudi Arabia, acts as a prime example of urban expansion and its related challenges [4]. Observing a remarkable 225% growth in urbanization over four decades, Najran leads in the rapid development of urban areas. It functions as the administrative center and the most populated city in the Najran region, covering an area of 578.64 square kilometers and housing nearly two-thirds of the region's population, with 454,035 of the roughly 665,031 people in 2019.

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The significant increase in the number of vehicles, rising from 94,403 in 2000 to about 240,000 by the end of 2019, has pressured the urban road network and contributed to a surge in RTAs. With 1,140 transportation-related accidents recorded in 2019, which account for nearly half of all traffic accidents in the Najran region, it's evident that addressing these issues is crucial [5], [6].

This study initiates a crucial assessment of the spatial dynamics of RTAs in Najran, at the crossroads of rapid urbanization and increased vehicular demand. It aims to explore the complex relationship between traffic accidents and the urban landscape, guided by key questions:

i. Is there a clear correlation between traffic accidents and the types of roads in Najran?
ii. Are road traffic accidents more frequent in the city's densely populated districts?
iii. Where are the accident hotspots located within the city?

Several studies on RTAs have provided a comprehensive assessment of their spatial distribution, underlying causes, and temporal dynamics, drawing insights from a multitude of geographic contexts. For example, Geographical Information Systems (GIS) and Kernel Density Estimation (KDE) were employed in a study to assess the spatial patterns of injury-related RTAs in London, UK [7]. Another study employed an integrated KDE algorithm and spatial auto-correlation analysis to determine RTAs hot spots and simultaneously evaluate the statistical significance of the hot spot clusters [8]. One another study integrated network kernel density estimation (NetKDE) with local Moran'I for hot spot detection of RTAs [9]. A study in Dhaka, Bangladesh identified hot spot zones using Local Moran's-I Statistic, Getis-Ord Gi* statistic and KDE and examine spatial patterns of high cluster of RTAs [10]. Furthermore, a study employed clustering analysis to have a better understanding of RTA patterns in complex urban network of Mashhad city, Iran. They employed traditional KDE, nearest neighbor analysis and K-function [11]. Similarly, in Indian Gurgaon Jaipur National Highway (NH-8), based on the data from 2010 to 2012, KDE, Moran's I statics and Get-Ord Gi* Statics are were used for the spatial distribution of RTAs [12]. In Turkey Afyonkarahisar and Konya provinces, spatial analysis approaches including Moran's I, GetisOrd G and planar and NetKDE were employed to assess the clusters of RTAs [13].

The various districts of Jeddah. The GWPR was also employed to assess the correlation between the various pedestrian facilities and the factors related to land use and transport infrastructure [17]. The GIS-based assessments of vehicular speed at GCC and King Fahd highways in Eastern Province of Saudi Arabia revealed high percentage of RTAs at high-speed zones [18].

Despite existing research on RTAs in various cities across Saudi Arabia, there remains a critical gap in studies specifically targeting Najran city. Addressing this gap, conducting comprehensive research in Najran not only holds the potential to enhance our understanding of the unique spatial patterns of RTAs within the city but also provides a foundation for developing targeted interventions. By analyzing RTAs hotspots within Najran, this study will present valuable insights, thereby enabling local authorities to implement effective road safety measures. Figure 1 illustrates our proposed study framework.

The study can be summarized as follows:

- Utilized Density-based Spatial Clustering of Applications with Noise (DBSCAN) and Hierarchical Agglomerative Clustering (HAC) to identify accident hotspots. Applied Kernel Density Estimation (KDE) to assess the concentration of accidents. Conducted Point Density Analysis to identify high-density areas. Used Proximity Analysis with Thiessen polygons to evaluate spatial distribution and coverage of essential services.
- Implemented Spatial Interpolation Analysis, including Inverse Distance Weighting (IDW) and Kriging, to predict accident occurrence in unobserved areas.
- Mapped the spatial distribution of road traffic accidents, pinpointing critical hotspots along major highways and arterial roads. Assessed the accessibility and coverage

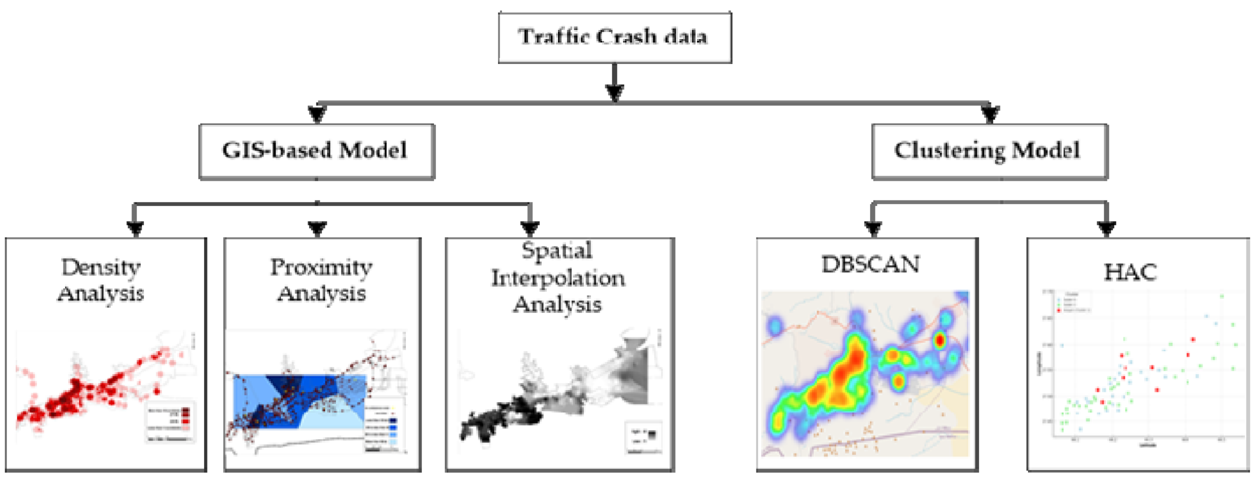


FIGURE 1. Flow chart of our proposed study.

of essential services like ambulances and hospitals in relation to accident sites.
- Provided insights to urban planners and local authorities for enhancing road safety measures. Provided recommendations for optimizing emergency response times and resource allocation.

II. MATERIALS AND METHODS

A. STUDY AREA

Najran is situated in Saudi Arabia's extreme southwest spanning latitudes between 17°44′17″ and 24°17″ and longitudes from 7°45′ to 41°26′ at an altitude of 1210 meters above sea level. It presents a unique area for study due to its distinct geographical and administrative features. Figure 2 illustrates the administrative boundaries and geographic setting of Najran setting the stage for an in-depth exploration of traffic accident trends within the city. This study seeks to map out the spatial patterns of traffic accidents in Najran with the goal of uncovering trends that can contribute to improvements in urban safety and planning strategies, especially considering the city's fast-paced urbanization and growth in infrastructure.

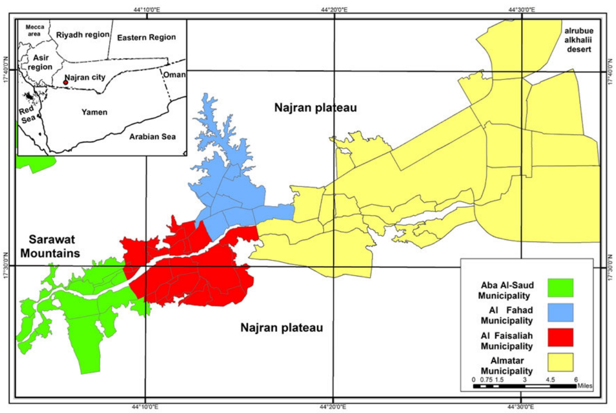


FIGURE 2. Administrative boundaries of Najran city, Saudi Arabia.

This study of spatial patterns of traffic accidents within the city of Najran, we employ Clustering Algorithms including Density-based Spatial Clustering of Applications with Noise (DBSCAN) [19] and Hierarchical Agglomerative Clustering (HAC) [20] as well as advanced Geographic Information Systems (GIS) techniques to analyze the RTAs hot spots.The GIS-based strategies encompasses two principal density analysis methods: Kernel Density Estimation (KDE) and Point Density Analysis, alongside Proximity Analysis using Thiessen polygons, and Spatial Interpolation Analysis, which includes both Inverse Distance Weighting (IDW) and Kriging. These techniques are instrumental in elucidating the spatial distributions of RTAs, providing a comprehensive understanding of their patterns and implications for urban planning and public safety initiatives.

B. CLUSTERING ALGORITHMS

1) DENSITY-BASED SPATIAL CLUSTERING OF APPLICATIONS WITH NOISE (DBSCAN)

It is a popular clustering algorithm known for its ability to find arbitrarily shaped clusters and its robustness to outliers. It operates based on the idea of identifying 'core' samples within dense areas of a dataset and expanding clusters from them. The key components involved in DBSCAN are Core Points, Border Points, and Noise. A point is considered as a core point if at least a minimum number of points (*MinPts*) are within a given distance (ε) from it. The ε is the distance threshold within which points are considered to be within the same neighborhood. It determines the radius of the neighborhood around a point. Mathematically, for a point p to be a core point, the condition must be satisfied as shown in Eq. 1

$$|\{q \in D | dist\ (p, q) \leq \varepsilon\}| \geq MinPts \tag{1}$$

where,

D is the dataset, and $dist\ (p, q)$ is the distance between points p and q.

Similarly, a point is a border point if it is not a core point but is within the ε distance of a core point. A point is considered noise if it is neither a core point nor a border point. The algorithm involves following steps:

- For each point in the dataset, count how many points fall within the ε-distance. If a point has at least *MinPts* within this distance, mark it as a core point.
- For each core point, if it is not already assigned to a cluster, create a new cluster. Then recursively add all directly reachable points to this cluster. Directly reachable means being within ε distance from the core point and having sufficient *MinPts*in their neighborhood.
- Assign border points to one of the clusters of their associated core points.
- Any point that is not a core point or a border point is considered noise and does not belong to any cluster.

2) HIERARCHICAL AGGLOMERATIVE CLUSTERING (HAC)

HAC HAC operates by progressively merging clusters starting from individual data points, effectively building a hierarchy from the bottom up. Initially, each point in the dataset is considered a separate cluster. At each step of the algorithm, the two clusters that are closest according to a chosen distance metric are merged. This process repeats until all points belong to a single cluster or a predetermined number of clusters is reached. The closeness between clusters can be defined in several ways, including Single Linkage, where the distance between two clusters C_i and C_j is the minimum distance between any point in C_i and any point in C_j given by Eq. 2.

$$d\left(C_i, C_j\right) = \min\left\{dist(a, b) : a_i \in C_i, b_j \in C_j\right\} \tag{2}$$

Another approach, Ward's Method, minimizes the total within-cluster variance, with the increase in squared error after merging two clusters C_i and C_j simplified to ΔSSE as shown by Eq. 3.

$$\Delta SSE = \left\| \mu_{c_i \cup c_j} - \mu_{c_i} \right\|^2 + \left\| \mu_{c_i \cup c_j} - \mu_{c_j} \right\|^2 \tag{3}$$

where, μ_{c_i} and μ_{c_j} are the centroids of the original clusters, and $\mu_{c_i \cup c_j}$ is the centroid of the merged cluster.

The iterative process of merging in HAC and its results are often visualized using a dendrogram, which reveals the sequence and distance at which each merger occurs, assisting in the decision of the final number of clusters. The dendrogram highlights significant jumps in distance, indicating natural separations in the data. Ward's Method, in particular, focuses on creating compact, spherical clusters by choosing merges that minimize the increase in total squared error, ensuring clusters are as homogeneous as possible. This method's goal to minimize ΔSSE leads to a balanced distribution of cluster sizes and shapes, making it a favored choice for many applications that require clear delineation of cluster boundaries based on internal variance.

C. GIS-BASED APPROACHES

1) DENSITY ANALYSIS

Density analysis is crucial for understanding the distribution and concentration of spatial phenomena, such as traffic accidents, across a geographical area. Two common methods for conducting density analysis are Kernel Density Estimation (KDE) [21] and Point Density Analysis [21]. Both approaches provides insights into the density of events, but they differ in their methodology and underlying mathematical principles. Kernel Density Estimation is a non-parametric way to estimate the probability density function of a random variable. In the context of spatial analysis, KDE is used to create a smooth surface (or heatmap) that represents the density of point features (e.g., locations of traffic accidents) across the area.

Given a set of points $X = \{x_1, x_2, x_3, \ldots, x_n\}$ in a two-dimensional space, the kernel density estimate at location $x$is given by Eq. 4.

$$\hat{f}(x) = \frac{1}{nh^2} \sum\nolimits_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{4}$$

where:

- n is the number of points,
- h is the bandwidth (a user-defined parameter that controls the level of smoothing),
- K is the kernel function (a symmetric function that integrates to one, often a Gaussian function).

The choice of bandwidth h is critical in KDE, as it influences the degree of smoothing: a smaller results in less smoothing (and potentially more noise), while a larger h produces a smoother surface that may obscure local variations. The Point Density Analysis calculates the density of point features within a moving window that passes over the area of interest. This method is more straightforward than KDE and results in a raster grid where each cell's value represents the density of points within the surrounding neighborhood. The density at a grid cell $\Gamma(c)$ is computed by using Eq. 5.

$$\Gamma(c) = \frac{\eta_p}{\alpha} \tag{5}$$

where:

- η_p is the number of points within window
- α is the area of window

The window can be a circle (with a specified radius) or a square (with a specified side length), and the area of the window depends on its shape and size. This method is sensitive to the chosen size of the moving window, as a larger window will encompass more points, leading to a lower resolution of the density surface.

2) PROXIMITY ANALYSIS

Proximity analysis using Thiessen polygons, also referred to as Voronoi diagrams, is a powerful technique for spatially delineating areas of influence around specific points [22]. This method is particularly useful in GIS for a variety of applications, including urban planning, environmental management, and, specifically, analyzing spatial patterns related to traffic accidents. Thiessen polygons help in understanding how geographical proximity to certain features, like accident sites, influences surrounding areas [23], [24].

Thiessen polygons are created by first taking a set of seed points in a plane. For each point, a polygon is constructed such that any location within the polygon is closer to its corresponding seed point than to any other seed point. The mathematical expression for the Voronoi cell (Thiessen polygon) associated with a point in a set of points $P = \{p_1, p_2, p_3, \ldots, p_n\}$ can be defined as Eq. 6.

$$V(p_i) = \left\{ x \in \Re^2 | d(x, p_i) \leq d(x, p_i)\, \forall j \neq i \right\} \tag{6}$$

where $V(p_i)$ represents the Voronoi cell for point p_i, x is any point in the plane, and $d(x, p)$ denotes the Euclidean distance between x and p.

The basic steps are as follow:

- Identify Seed Points: In a traffic analysis scenario, these could be locations of traffic accidents.
- Calculate Perpendicular Bisectors: For every pair of seed points, calculate the perpendicular bisector of the segment connecting them. These bisectors are potential edges of the Voronoi cells.
- Intersect Bisectors: The intersection points of these bisectors form the vertices of the Voronoi cells.
- Construct Polygons: The bounded regions formed around each seed point, delineated by the bisectors, are the Thiessen polygons. Each polygon encompasses the area closer to its seed point than to any other.

D. SPATIAL INTERPOLATION ANALYSIS

Spatial interpolation predicts accident occurrences at unsampled locations based on patterns observed in the locations of known accidents. Two widely used methods for spatial interpolation are Inverse Distance Weighting (IDW) [25] and

Kriging [26]. The IDW is based on the premise that the value at an unsampled point can be approximated as a weighted average of the values from nearby sampled points. The weight decreases as the distance between the unsampled point and the sampled points increases, implying that nearer observations have a stronger influence than those further away [27]. This method is straightforward and does not require the data to follow a specific statistical distribution [26]. The general formula for IDW is given by Eq. 7.

$$z(x) = \frac{\sum_{i=1}^{n} w_i \cdot z_i}{\sum_{i=1}^{n} w_i} \tag{7}$$

where,

- $z(x)$ is the predicted value at the unsampled location x.
- z_i are the known values at the sampled locations.
- w_i are the weight assigned to each sampled value, calculated as $1/d_i^p$, where d_i is the distance between the unsampled location x and each of the sampled locations, and p is a power parameter that controls how quickly the influence of a point decreases with distance.

The Kriging is a more sophisticated method that not only considers the distance between sampled and unsampled points but also incorporates the spatial auto-correlation between observed values into the prediction [28]. This method models the variance and covariance of the data points, allowing for more accurate and statistically robust predictions. Kriging is especially valuable when the spatial relationship between data points is complex and non-linear [29]. The Kriging prediction is formulated as Eq. 8.

$$z^{*}(x_0) = \sum_{i=1}^{n} \lambda_i \cdot z(x_i) \tag{8}$$

where,

- $z*(x_0)$ is the predicted value at the unsampled location x_0.
- $z(x_i)$ are the observed values at the sampled locations.
- λ_i are the weight weights calculated for each observed value such that the prediction error is minimized, subject to unbiasedness constraints. These weights are derived from the spatial auto-correlation structure of the data, which is typically modeled using a semivariogram.

III. RESULT AND DISCUSSION

In this study, the distribution and impact of RTAs within the urban landscape of Najran city are analyzed through an extensive dataset, revealing significant spatial variations across the city's districts. The data, derived from official records for the year 2022, encompasses the numerical distribution and percentage of traffic accidents in different districts. Our analysis was conducted in 2019 based on the most recent data available at the time, which was the 2019 dataset. Moreover, the authors have updated the dataset to include 2022 data, and they observed that the differences between the 2022 and 2023 datasets are minimal and unlikely to impact the analysis results significantly. A dataset for 2023 was not publicly available. Therefore, this analysis examined trends over a specific period ending in 2022. It is most appropriate to use the 2022 dataset for this purpose.

This comprehensive dataset is instrumental in identifying high-risk areas within the city and assessing the effectiveness of traffic safety measures over time. In addition, python programming environment was used for the implementation of clustering algorithms while the "Arc GIS 10.8" software was employed to explore the spatial characteristics. By integrating RTAs data with spatial layers like road networks, traffic density, land use, and demographics, GIS uncovers accident patterns and hot spots. A crucial step in this process was the creation of a geo-database using the "Arc Catalogue" tool, which compiled various urban data into feature classes. This database enabled a comprehensive suite of spatial analyses, in-clouding density analysis, proximity analysis, and spatial interpolation analysis. It is also pertinent to mention two performance metrics including Silhouette Score [30] and Ca-linski-Harabasz Score [31] were employed to assess the efficient of DBSCAN and HAC. The Silhouette Score measures show similar an object is to its own cluster compared to other clusters. A higher score indicates better-defined clusters. The Calinski-Harabasz index is a variance-ratio criterion, where a higher score typically suggests that clusters are dense and well-separated.

A. DISTRIBUTION OF TRAFFIC ACCIDENTS

The distribution of traffic accidents in Najran city is characterized by two distinct spatial variables: the relative distribution across neighborhoods and the distribution based on road types.

1) RELATIVE DISTRIBUTION OF TRAFFIC ACCIDENTS IN NEIGHBORHOODS

The spatial pattern of traffic accidents within Najran's neighborhoods presents a variegated landscape of risk as shown in Figure 3. Some areas, especially in the eastern districts, emerge as significant clusters, experiencing a relative concentration of accidents surpassing 4% of the city's total. These high-incidence districts starkly contrast with the moderate-accident zones in the central and eastern regions, where accident prevalence hovers between 2% and less than 4%. The majority of neighborhoods, however, demonstrate a lower occurrence, with each district accounting for less than 2% of traffic accidents, indicating a widespread but diluted risk across the city. Notably, in the northeastern districts, a complete absence of reported accidents underscores regions of potential urban traffic safety success or lower traffic activity. This nuanced distribution underscores the complexity of traffic patterns and the necessity for targeted safety interventions in specific neighborhoods.

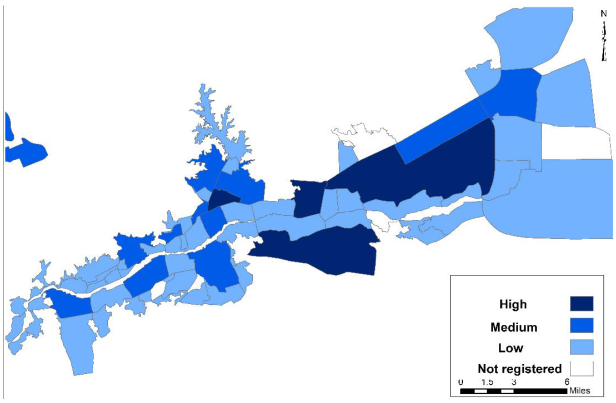


FIGURE 3. Relative distribution of traffic accidents in various districts of Najran city.

2) DISTRIBUTION OF TRAFFIC ACCIDENTS BASED ON ROAD TYPES

Analyzing the frequency of accidents according to road types reveals a compelling correlation between accident rates and the hierarchical structure of the road network as shown in Figure 4. Highways bear the majority of traffic incidents, reflecting their role as primary conduits for vehicular movement and the associated risk due to higher traffic volumes. Arterial roads follow, contributing a significant proportion of traffic accidents, with specific roads like King Abdul Aziz Road being particularly accident-prone due to their heavy usage. Collector roads, while less congested, still account for a noteworthy percentage of traffic mishaps. Local roads, in contrast, record the fewest accidents, which can be attributed to their comparatively lower traffic flow.

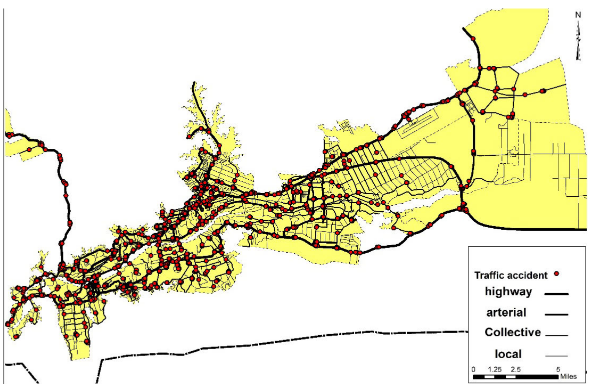


FIGURE 4. Distribution of traffic accidents on the bases of road type in various districts of Najran city.

B. HOT SPOT ASSESSMENT BY USING CLUSTERING ALGORITHMS

Figure 5 presents a DBSCAN-generated heatmap visualizing the spatial distribution of road traffic accidents within Najran City. The map uses color intensity to represent accident densities, with darker colors indicating higher concentrations of accidents in Cluster 0, which includes 42 locations, and lighter colors representing the 34 outlier locations with fewer accidents. The legend categorizes these intensities from low to high, aiding in the identification of both prominent accident hotspots and lesser-affected areas. This visual arrangement facilitates a clearer understanding of the geographic spread of accidents, highlighting critical areas for potential safety improvements and urban planning interventions.

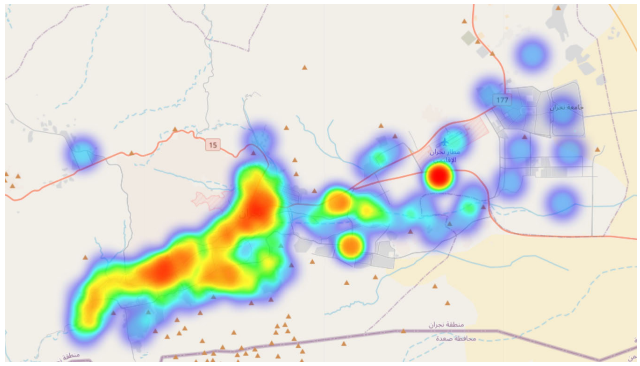


FIGURE 5. Hot spot heatmap generated via DBSCAN.

Figure 6 displays a scatter plot created using Hierarchical Agglomerative Clustering (HAC) to identify hotspots of road traffic accidents in Najran City. This plot delineates three distinct clusters based on the similarity in accident frequencies across different districts. Cluster 0, represented by light blue 'X' markers, and Cluster 2, marked with green 'X' markers, indicate areas with varying degrees of accident occurrences. The most critical areas, termed as hotspots and denoted by solid red circles, belong to Cluster 1. These red circles highlight districts with the highest frequency of accidents, signifying them as primary focus areas for traffic safety interventions. The effectiveness of HAC in this analysis is underscored by Silhouette and Calinski-Harabasz Scores of 0.564 and 233.73, respectively, which suggest that HAC provides a more meaningful and distinct classification of accident clusters compared to DBSCAN, thus offering clearer insights for urban safety planning.

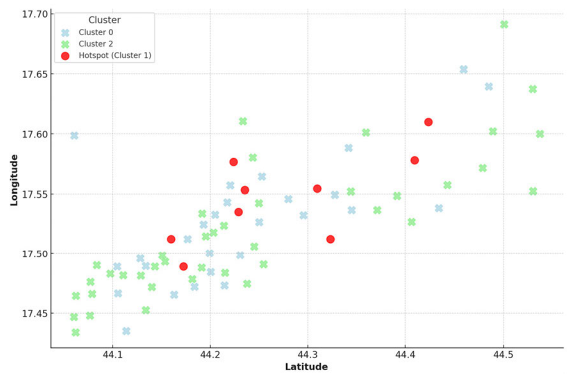


FIGURE 6. Hot spot heatmap generated via DBSCAN.

C. IDENTIFICATION OF HOT AND COLD SPOT BASED ON THE DENSITY ANALYSIS APPROACHES

The analysis conducted through ArcMap's Mapping Clusters tools reveals distinct patterns of traffic accident concentrations within Najran city, delineating hot and cold spots with a statistical backbone of confidence levels (90%, 95%, and 99%). This approach confirms previous findings and enriches our understanding of the spatial distribution of traffic accidents across the urban landscape. Figure 7 revealed that the traffic accident hotspots predominantly anchor in the western sectors of Najran, manifesting in two principal zones with a high statistical confidence. The first zone exhibits a horizontal expanse stretching from the Al Amlah district in the north to the Rajla district in the south. The second zone, in contrast, extends longitudinally from the Maqan district to the Hadhan district. These areas are particularly notable along the pivotal axes of King Abdulaziz, King Abdullah, and Prince Sultan Bin Al Aziz roads, illustrating their significance in urban mobility and accident concentration. The statistical confidence levels for these hotspots are recorded at 99% (with an additional three hotspots), 95%, and 90% respectively, highlighting a focused clustering of traffic accidents that could inform preventive measures and urban planning strategies.

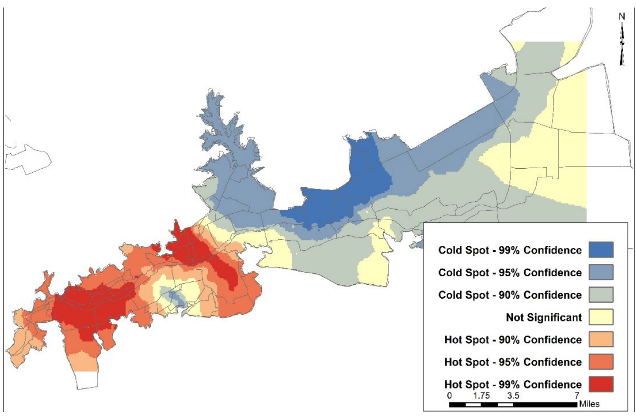


FIGURE 7. Hot and cold spot of traffic accidents.

Conversely, traffic accident cold spots gravitate towards the northern locales of Najran, particularly in the Al-Senaiyyah, Orissa, parts of Al Ghuwayla, Aba Elkhirit, and Al Massamah districts, achieving the highest negative value at a 99% confidence level. This distribution shows a lesser frequency of traffic accidents, potentially due to lower population density or different urban configurations. The analysis also notes cold spots dispersing towards the western and eastern parts of the aforementioned zone and predominantly in the city's southern sections at a 90% confidence level, indicating varied urban dynamics that mitigate accident occurrences. It is crucial to acknowledge areas within downtown Najran and its western and eastern parts where the G value is zero, indicating no significant statistical evidence of traffic accident distribution. This neutrality shows that accidents in these areas occur randomly and are not influenced by identifiable urban factors, challenging the notion of predictable patterns and highlighting the complexity of urban traffic dynamics.

1) HEATMAP OF TRAFFIC ACCIDENTS BASED ON DENSITY POINT ANALYSIS

The point density analysis, visualized in Figure 8, illustrates the frequency of traffic accidents across different regions of Najran, revealing critical patterns of high, moderate, and low accident occurrences. This spatial assessment is instrumental in identifying the urban areas most impacted by traffic accidents, thereby informing targeted safety measures and infrastructural improvements. The analysis pinpoints two primary regions within Najran that exhibit a high density of traffic accidents, each recording more than ten incidents. The downtown area, recognized as the commercial nucleus of the city, emerges as a significant hot spot. Here, the confluence of major roads creates a complex traffic dynamic prone to accidents. Similarly, the eastern part of the city along King Abdulaziz Road is identified as another high-density accident zone. These areas, characterized by their important roles in urban traffic flow and economic activity, necessitate focused attention for traffic safety and urban planning.

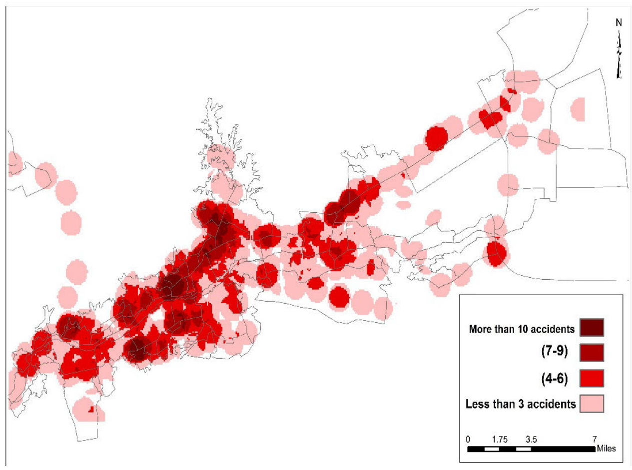


FIGURE 8. Heatmap generated by density point analysis.

Intersections across Najran's road network are marked by a pronounced increase in traffic accidents, with figures ranging from seven to nine incidents. This trend illustrates the intersections' role as critical points of traffic conflict and congestion. Additionally, the areas of moderate accident intensity, particularly at the junctures of arterial and collector roads, where accident numbers fluctuate between four to six are also highlighted. The analysis reveals that the combination of narrow roadways, high population density, and irregular vehicle patterns contribute to this moderate risk level, emphasizing the need for infrastructural adjustments and enhanced traffic management.

Conversely, certain locales, especially along the El Magharra road in the western parts of the city, are noted for their relatively low occurrence of traffic accidents, with three or fewer incidents. This dispersion of low-density accident areas across the city, particularly in less central regions, might reflect varied factors such as lower traffic volumes, better road conditions, or effective local traffic controls.

2) HEATMAP OF TRAFFIC ACCIDENTS BASED ON KDE ANALYSIS APPROACH

The KDE analysis reveals that high-density accident zones, defined by 36 accidents or more, predominantly align with the city's major highways as shown in Figure 9. These areas follow a longitudinal axis stretching from the west to the east of Najran, including horizontal roads that serve as critical connectors between these main thoroughfares. The concentration of high-density zones along these routes underscores their significance in the urban traffic network, highlighting them as focal points for traffic flow and, consequently, accident occurrences.

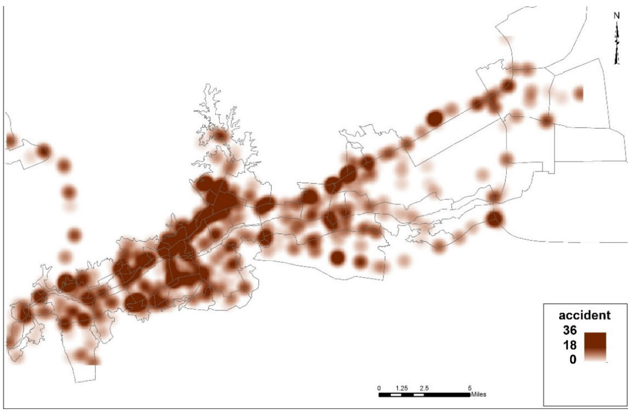


FIGURE 9. Heatmap generated by KDE analysis.

In contrast, areas characterized by a moderate density of traffic accidents, quantified as 18 accidents, are primarily concentrated around the city's main intersections. These intersections represent crucial nodes within Najran's road network, where vehicular paths converge and diverge, increasing the likelihood of traffic conflicts and accidents. The identification of these zones as areas of moderate accident density illustrates the need for targeted safety measures, such as improved traffic signalization, enhanced visibility, and perhaps the redesign of intersections to reduce conflict points.

D. PROXIMITY ANALYSIS: SITE ALLOCATION FOR HOSPITAL AND AMBULANCE SERVICE

The study conducted a proximity analysis to understand the distribution and impact of ambulance stations and public hospitals in the city of Najran in relation to traffic accidents. This analysis utilized Thiessen polygons to determine the allocated space for healthcare facilities and correlate it with the incidence of accidents within their respective service areas.

1) AMBULANCE SERVICES ANALYSIS

Najran is equipped with 11 ambulance stations spread across the city from west to east. The analysis, illustrated in Figure 10a, revealed a total of 1,018 traffic accidents within the areas served by these ambulance stations. Interestingly, 122 accidents occurred outside the coverage areas of these stations, indicating potential gaps in service coverage. The distribution of ambulance stations varies significantly in terms of service area size and the number of accidents covered:

- One station in the Al Shorfa district covers an extensive area of over 90 km^2 and accounted for 73 accidents, highlighting concerns over response times.
- Stations serving areas between 60 to 90 km^2 experienced 303 traffic accidents, with the Al Faysaliah station being particularly noteworthy.
- The highest accident rate, involving 431 incidents, was observed in stations covering 30 to 60 km^2, primarily located in the downtown and western parts of the city.
- Stations with service areas under 30 km^2 accounted for 211 accidents, also concentrated in downtown and western Najran.

2) PUBLIC HOSPITALS ANALYSIS

Najran hosts six public hospitals, mainly centralized within the city, covering 210 km^2 or 36.3% of the city's total area. This segment experienced 603 RTAs, whereas 537 accidents occurred beyond the hospitals' service areas, again highlighting potential delays in emergency services as illustrated by Figure 10b.

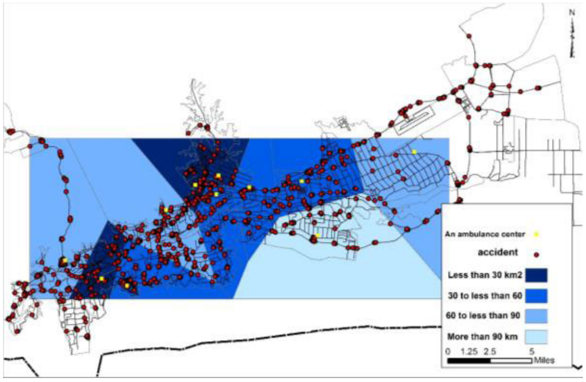
Spatial analysis further detailed the distribution of hospital spaces:

- Najran Old General Hospital, located in the Aba Al-Saud district, occupies the largest area at approximately 51 km^2, with 143 reported accidents.
- The New Public Hospital in Al Shariqa district follows, covering 46 km^2.
- The Maternity and Child Hospitals (MCH), King Khaled Hospital, and the Military Hospital each have similar spaces around 36 km^2 but differ in accident numbers: 111, 119, and 132 accidents, respectively.
- The University Hospital has the smallest allocated area of 3.9 km^2, correlating with the lowest accident number of 45.

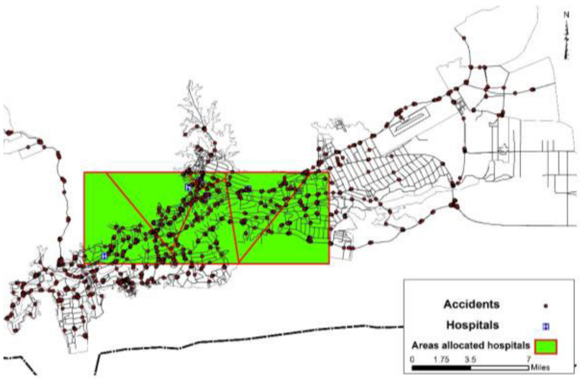
These findings shows the critical need for strategic planning in the placement of healthcare facilities and emergency services to optimize coverage and response times, thus potentially reducing the impact of RTAs in Najran. Figures 10a and 10b, which graphically represent the data, play a crucial role in visualizing the spatial distribution and areas of service provided by these essential public health infrastructures.

E. SPATIAL INTERPOLATION

The study on spatial interpolation has effectively transformed discrete, non-spatial point features into a comprehensive



(a)



(b)

FIGURE 10. Proximity analysis for Najran city; (a) Ambulance service allocation analysis; (b) Hospital allocation analysis.

surface map, illustrating the spatial dynamics of RTAs within the study area. By utilizing both Inverse Distance Weighted (IDW) and Kriging methods for their renowned precision, we have shed light on critical insights into the distribution of accidents across the city. Based on the Inverse Distance Weighted (IDW) analysis, Figure 11 displays the concentration zones, intermediate accident zones, and low accident areas. In terms of concentration zones, there are distinct areas, especially in downtown and the western sectors of the city, where RTAs have surged (more than four accidents), constituting roughly one-tenth of the city's total space.

These areas are categorized into three primary zones: a central corridor extending from north to south, an 'M'-shaped area in the south, and several western areas, including the Dahda district to the northwest of downtown. Regarding intermediate accident zones, regions with two to three accidents, covering 35.1% of the city's space, flank the high-accident zones. These include newly identified zones in the southern downtown and eastern horizontal sections. Likewise, the low accident areas constitute the majority of the city, where over half of its total area experiences two or fewer accidents. These zones stretch longitudinally from the northern outskirts through the central districts to the eastern parts, indicating a broad yet sparse distribution of road traffic incidents.

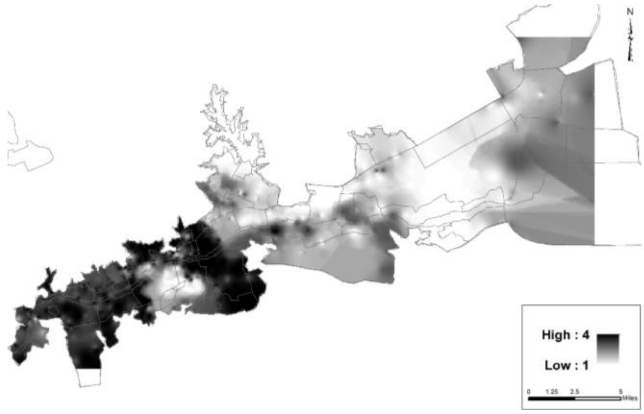


FIGURE 11. Spatial interpolation based on IDW.

The Kriging analysis (Figure 12) further emphasizes the western city parts as critical areas for accidents, where three to four incidents are more likely due to high population density and significant commercial activities. The dispersed accident zones are the areas experiencing two to three accidents are dispersed across downtown, western, and eastern city parts, showcasing a diverse geographic spread of RTAs. Remarkably, zones with only one accident account for 37.6% of the city's total space, indicating that such incidents are widespread. This distribution illustrates a higher accident intensity in the northern and eastern parts, attributed to dense agricultural activities in the east and the connecting desert roads.

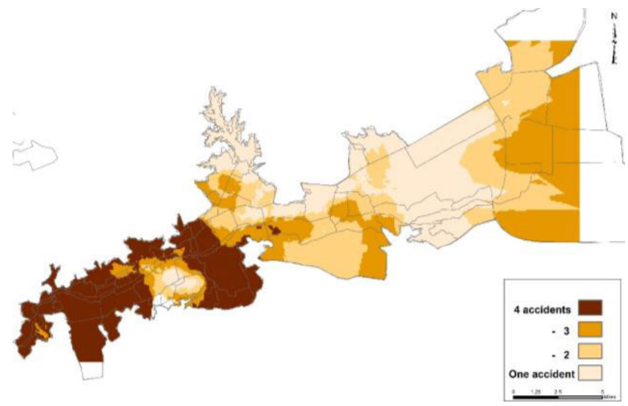


FIGURE 12. Spatial interpolation based on Kriging analysis.

IV. CONCLUSION AND RECOMMENDATIONS

This study utilized advanced clustering and GIS techniques to map the spatial patterns of road traffic accidents (RTAs) in Najran. Findings indicate that RTAs are concentrated along major highways like King Abdul Aziz and King Abdullah Roads, particularly in densely populated districts such as Al Khalidiyyah and Al Shorfa. Hierarchical Agglomerative Clustering (HAC) was more effective than DBSCAN in identifying significant clusters of accidents, as supported by higher Silhouette and Calinski-Harabasz Scores.

The analysis revealed that essential services such as ambulance stations and public hospitals are not optimally located relative to the accident hotspots, which underscores a need for strategic reallocation to enhance emergency response capabilities. Additionally, the study highlights the impact of road types on accident frequencies, suggesting that arterial roads, because of their heavy traffic and complex design, are particularly hazardous.

In response to these insights, we recommend a multifaceted approach to road safety in Najran. This should include redesigning traffic infrastructure to focus on accident hotspots, improving emergency service accessibility, and enhancing public awareness on safe driving practices. Embracing technology in traffic monitoring and predictive analytics will also be crucial in refining safety strategies.

As future directions, our initial study on the spatial analysis of RTAs in Najran utilized traditional clustering algorithms such as DBSCAN and Hierarchical Agglomerative Clustering (HAC) due to their proven effectiveness in identifying patterns and hotspots in geospatial data. However, recognizing the potential advantages of deep learning in handling more complex datasets, we are considering the exploration of deep clustering algorithms for future studies such as Deep Embedded Clustering (DEC) and Deep Clustering Network (DCN).

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